Day 8.

Block-1 APTITUDE (Mixtures & Averages).

1) The average of 5 numbers is 20. Find the total sum.

Sol:- given
Average = 20,
No. of values = 5

Sum = Average × Number

= 20 × 5

= 100

Answer: 100

2) The average of 10 number is 15. If one number is removed, the average becomes 14. Find the removed number.

Sol: given
Average of 10 numbers = 15
Total sum = 10 × 15 => 150

Average 14 for 9 number
New Sum = 9 × 14
= 126.

Removed number = 150 - 126

= 24

Answer: 24

3. A container has milk and water in ratio 3:2. If 10 litres of water is added, ratio becomes 3:4. Find initial quantity.

Sol:- given

Ratio = 3:2

Milk = 3x, water = 2x

After adding 10 L water →

ratio → 3:4

Milk = 3x

water = 2x+10

$$\frac{3x}{2x+10} = \frac{3}{4}$$

→ 12x = 6x+30 ; ⇒ 12x - 6x = 30

6x = 30 x = 30/6

$\boxed{x = 5}$ → Answer.

4. Find average of first 20 natural numbers.

Sol:- given

$$\text{Average} = \frac{n+1}{2}$$

$$\text{Average} = \frac{(20+1)}{2}$$

$$= 21/2$$

$\boxed{= 10.5}$ → Answer.

5) The average age of 5 students is 18. A new student joins and average becomes 19. Find age of new student.

Sol:- Average of 5 students = 18

Total = 5 × 18

= 90

After new join students,

Average = 19 for 6 students

New total = 6 × 19

= 114

Age of new student = 114 - 90

= 24 Answer.

Block 2 - VERBAL ABILITY

Reading Passage

1) why are data structures important?

Data structures are important because it helps to organize and store data efficiently.

2) Name some operations performed on data.

⇒ Searching

⇒ Deleting

⇒ Inserting

3. How does choosing the right data structure help performance?

Choosing right data structure helps to improves the performance significantly.

GRAMMAR CORRECTION.

1) He don't understand data structure

He doesn't understand data structure.

2) She have completed the assignment

She has completed the assignment.

3) They was working on the project.

They were working on the project.

Block 3 - COMMUNICATION PRACTICE.

My favorite programming language is java because it easy to learn and powerful.

It is used in web development, mobile apps, and many software applications.

Java has features like platform independence and safe in security, which make it very useful.